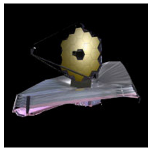

9320 - Are Wake Trails Ubiquitous in WR+BH HMXBs?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Photometric imaging timeseries (Phase coverage) of X-ray Binary				
	1	Phase 0.0	NIRCam Imaging	(1) CXOU-J005510.0-374212
	11	Re-observe missed mosaic dithers	NIRCam Imaging	(1) CXOU-J005510.0-374212
	2	Phase 0.1	NIRCam Imaging	(1) CXOU-J005510.0-374212
	3	Phase 0.2	NIRCam Imaging	(1) CXOU-J005510.0-374212
	4	Phase 0.3	NIRCam Imaging	(1) CXOU-J005510.0-374212
	5	Phase 0.4	NIRCam Imaging	(1) CXOU-J005510.0-374212
	6	Phase 0.5	NIRCam Imaging	(1) CXOU-J005510.0-374212
	7	Phase 0.6	NIRCam Imaging	(1) CXOU-J005510.0-374212
	8	Phase 0.7	NIRCam Imaging	(1) CXOU-J005510.0-374212
	9	Phase 0.8	NIRCam Imaging	(1) CXOU-J005510.0-374212
	10	Phase 0.9	NIRCam Imaging	(1) CXOU-J005510.0-374212

ABSTRACT

Black hole high mass X-ray binaries are among the leading candidates to be the progenitors of binary black holes. The WR systems have the highest RV amplitudes, yet their BH masses are questionable due to the effects of X-ray irradiation on the stellar wind, and newly identified contribution of the accretion disk hotspot to line-profiles. JWST can obtain the binary IR lightcurve which has a predicted exact phase offset with respect to the X-ray eclipse seen by Chandra. In IC 10 X-1 the IR dip predicted by hot-wind + cool wind models lags the X-ray dip; implying a trailing wake of recombining material flowing from the sheltered cool sector. We propose to repeat the experiment with the near-twin system NGC 300 X-1 using contemporaneous X-ray and IR observations.

OBSERVING DESCRIPTION

These observations will be using both the modules of NIRCam providing a field of view for monitoring a large chunk of the NGC 300 galaxy. In order to account for the module gaps in both horizontal and vertical directions, FULLBOX-5TIGHT dither pattern will be applied which is more efficient in terms of overhead time than FULL and ensures no gaps in the stacked observation. The primary target will be placed in one pf the full-overlap regions. The first visit to the target will execute 2 additional FULLBOX5TIGHT dithers to form a wide-field mosaic of the galaxy. This strategy mirrors that used for the archival JWST observation of IC 10. Only JWST can do this science, as the largest ground-based IR telescopes (e.g., Gemini) are not sensitive enough given that the rapid binary motion constrains the allowed exposure time to a few tens of minutes at most.

Exposure time: JWST’s highly sensitive imaging capability allows for very high SNR using very low exposures. To fulfill both the primary and secondary objectives, the SNR needs to be high enough to detect the variability of the eclipsed and un-eclipsed phases in each observation of X-1, and also provide a significantly higher limiting magnitude of the stacked image (all visits) than previous studies. The exposure time depends on the choice of readout method, number of groups, and integration in a single exposure. We have chosen to use the BRIGHT2 readout pattern, using 3 groups and 4 integrations for each dither (total 5 dithers).

NIRCAM ETC estimates that the SNR for NGC 300 X-1 (B= 22.5, continuum shape similar to spectral type A0) will be ~90 for SW @ 1.5 microns, and ~60 for LW @ 2.77 microns. This is enough to fulfill both science goals.

Feasibility of scheduling: This system has a very precisely constrained ephemeris: Porbit = 32.7921±0.0003 hr, epoch of mid eclipse = MJD 59791.51 ([8], updated by adding 1500 orbital periods) and adding an extra half a period to give Epoch of Phase Zero = JD 2459792.69 which will make it easy to schedule observations accurately at the requested phases. The binary orbital period is relatively small (34.9 hr), so the time constraint imposed by the phase requirement will be less significant.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Proposal 9320 - Observation 1 - Are Wake Trails Ubiquitous in WR+BH HMXBs?

Observation

Proposal 9320, Observation 1: Phase 0.0

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
3	1	10.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULLBOX	5TIGHT	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements	
	Phase 0 to 0.1 with period 32.7921 Hours and zero-phase 2459792.69 HJD Sequence Visits within 53.0 Days Visits Same PA Offset 5.0 arcsec, -10.0 arcsec Fiducial Point Override NRCAS_FULL Same V3 PA 1, 11

Observation

Proposal 9320, Observation 11: Re-observe missed mosaic dithers

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: Obs 11 is a re-observation attempt for skipped dither positions in the Mosaic (Obs 1). Mosaic was taken at PA = 35.4. Hence to fill the gaps will require matching the PA of the original dithers.

No phase or time constraints apply to Obs 11.

Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 11:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULLBOX	5TIGHT	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements

Sequence Visits within 53.0 Days
Visits Same PA
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Same V3 PA 1, 11

Observation

Proposal 9320, Observation 2: Phase 0.1

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 2:1) Warning (Form): Data Excess over lower threshold

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	5	4	20	5	2308.406	196772

Special Requirements

Phase 0.1 to 0.2 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Observation

Proposal 9320, Observation 3: Phase 0.2

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements

Phase 0.2 to 0.3 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Observation

Proposal 9320, Observation 4: Phase 0.3

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 4:1) Warning (Form): Data Excess over lower threshold

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	5	4	20	5	2308.406	196772

Special Requirements

Phase 0.3 to 0.4 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Observation

Proposal 9320, Observation 5: Phase 0.4

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements

Phase 0.4 to 0.5 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Observation

Proposal 9320, Observation 6: Phase 0.5

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements

Phase 0.5 to 0.6 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Observation

Proposal 9320, Observation 7: Phase 0.6

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements

Phase 0.6 to 0.7 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Observation

Proposal 9320, Observation 8: Phase 0.7

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements

Phase 0.7 to 0.8 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Observation

Proposal 9320, Observation 9: Phase 0.8

Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	196772

Special Requirements

Phase 0.8 to 0.9 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL

Proposal 9320 - Observation 10 - Are Wake Trails Ubiquitous in WR+BH HMXBs?

Observation

Proposal 9320, Observation 10: Phase 0.9 Wed Dec 31 19:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CXOU-J005510.0-374212	RA: 00 55 9.9900 (13.7916250d) Dec: -37 42 12.16 (-37.70338d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Wolf-Rayet stars, X-ray binary stars]
Extended=NO

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	5	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	BRIGHT2	4	4	20	5	1878.935	

Special Requirements

Phase 0.9 to 1.0 with period 32.7921 Hours and zero-phase 2459792.69 HJD
Offset 5.0 arcsec, -10.0 arcsec
Fiducial Point Override NRCAS_FULL